

uk | SRC

SKAO Regional Centre United Kingdom

AI coding agents from prompts to loops

Dave
Morris
Manchester
University

November 2026

Dave Morris
dave.morris@manchester.ac.uk

Levels of AI adoption

Where we started in December 2025

Prompt Engineering

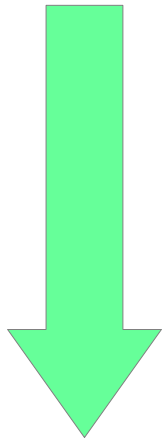
Crafting a single best instruction for an AI model

Context Engineering

Managing what information the model sees

Harness Engineering

Building the local environment where an agent runs



Where we are in October 2026

Loop Engineering

Designing workflows that drive the agent toward completion

https://www.ibm.com/think/topics/loop-engineering

Dave Morris
dave.morris@manchester.ac.uk

January 2026 - when it started

My ✨ WOW ✨ moment

This was when I realized what AI agents could do.

Cursor with Claude

Coding agent for building ambitious software.



https://cursor.com/



https://claude.ai/

Dave Morris
dave.morris@manchester.ac.uk

May 2026 - monthly susbscription

Cursor with Claude

CURSOR
https://cursor.com/

Claude
https://claude.ai/

Harness Engineering, automated attribution.

Monthly subscription + extras > $100/mth

```
AIMetrics: [
    {
    "timestamp": "2026-08-26T13:00:15",
    "name": "Cursor CLI",
    "version": "2026.02.13-41ac335",
    "model": "Claude 4.6 Opus (Thinking)",
    ....
    }
]
```


Dave Morris
dave.morris@manchester.ac.uk

June 2026 - this happened

June 12, 2026

The U.S. Department of Commerce imposed export controls on Anthropic's Fable 5 and Mythos 5 models restricting access to the models for any foreign national.



vs



“any foreign national” - that’s me !

Dave Morris
dave.morris@manchester.ac.uk

June 2026 - then this happened

June 16, 2026

xAI acquired Cursor's parent company, Anysphere, on August 14, 2026



CURSOR
https://cursor.com/

Sorry, but we can’t trust this guy ❌

Dave Morris
dave.morris@manchester.ac.uk

July 2026 - looking for alternatives

Avoid dependency on a single source

Avoid vendor lock-in

Looking for alternatives was already on my todo list.

Events just pushed it to the top of the list.

Todo list :

1. Evaluate open weights models

Dave Morris
dave.morris@manchester.ac.uk

August 2026 - DeepSeek in the news

August 13, 2026 - DeepSeek released the initial MIT-licensed open-source code on GitHub and entered public beta.

It broke viral developer records, gaining over 95,000 GitHub stars within its first 48 hours.

https://github.com/deepseek-ai/deepseek-harness

FOD#163: DeepSeek is having its second DeepSeek moment

https://www.turingpost.com/p/deepseek-harness-explained



AI Newsletter for Engineers and Researchers

Dave Morris
dave.morris@manchester.ac.uk

August 2026 - DeepSeek evaluation

deepseek
https://deepseek.ai/

Subjective comparison :

DeepSeek Harness is equivalent to Cursor with Claude Opus

```
AIMetrics: [
    {
    "timestamp": "2026-08-27T11:22:00",
    "name": "@deepseek-ai/dsh",
    "version": "0.1.1-rc.2",
    "model": "deepseek-v4-flash",
    ....
    }
]
```



Dave Morris
dave.morris@manchester.ac.uk

August 2026 - containerized agent

DeepSeek Harness inside a Docker container.

https://github.com/Zarquan/Lithosia-docker

Using volume mounts to control access to the host file system.

```
podman run \
    ....
    --volume "${PROJECT_PATH}:/Workspace" \
    ....
```

This is not just for DeepSeek, I ran Cursor in a container too.

Pro tip - don't give an AI agent unrestricted access to your home directory



Dave Morris
dave.morris@manchester.ac.uk

uk | SRC

SKAO Regional Centre United Kingdom

October 2026 - token cost

deepseek

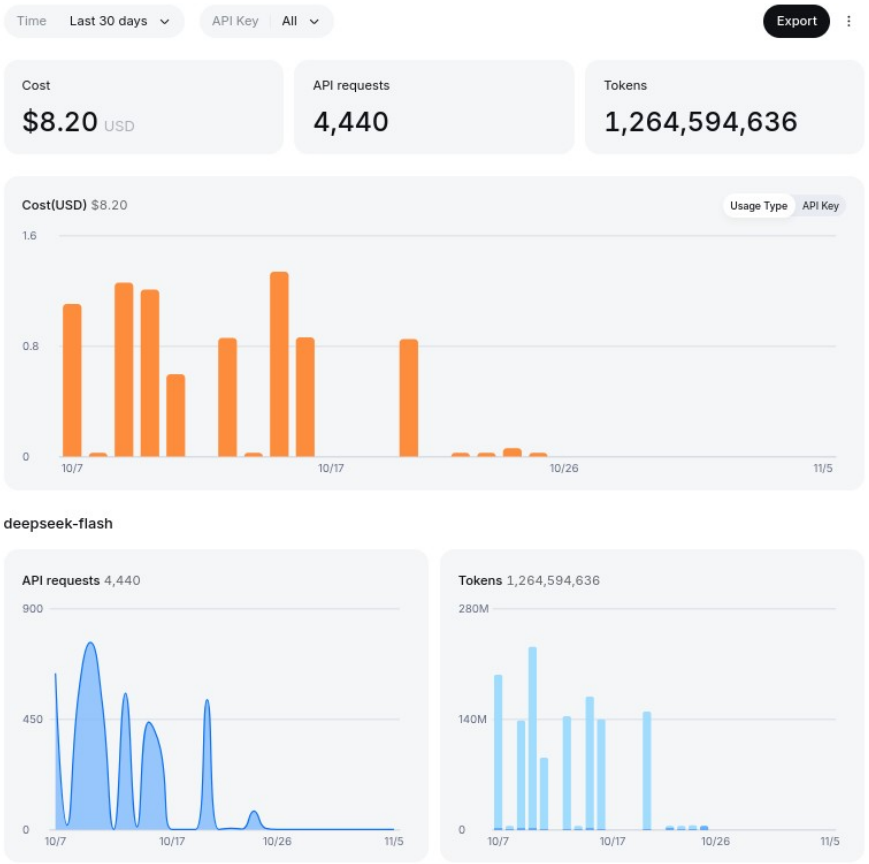
https://deepseek.ai/

Pay as you go API access :

1,264,594,636 tokens

4,440 requests

$8.20 USD



Dave Morris
dave.morris@manchester.ac.uk

October 2026 - benchmarks

deepseek
https://deepseek.ai/

LiveBench

Contamination-free long-horizon AI benchmark.

https://arxiv.org/abs/2406.19314
https://livebench.ai/

arXiv

#9 overall score

#1 Claude Fable 5.1	85.7	$7.406
#2 GPT-6 Astra	85.6	$4.828
...		
#9 DeepSeek V4.1 Flash 1017	72.9	$0.185

#3 on coding

#1 Claude Fable 5.1	97.7	$5.407
#2 Claude Opus 5.5	96.3	$5.224
#3 DeepSeek V4.1 Flash 1017	96.0	$0.204

Dave Morris
dave.morris@manchester.ac.uk

October 2026 - environmental impact

GreenPT
https://www.greenpt.ai/

GreenPT - European infrastructure
100% renewable energy

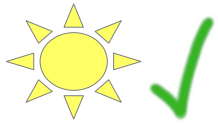
“Our servers run in Europe on renewable energy, with air cooling that uses a fraction of the water most data centers need. Lower footprint, by design, not by offset.”

“PUE (Power Usage Effectiveness) measures how efficiently a data center uses its electricity; Our infrastructure operates at a PUE of 1.25, compared to the industry average of 1.55.”

“WUE (Water Usage Effectiveness) measures the amount of water a data center consumes per unit of computing it delivers. Our hosting partner operates at a WUE of 0.25 L/kWh against an industry average of 1.8 L/kWh, an 86.1% reduction.”

Open weight models 

European infrastructure 

Renewable energy 

Dave Morris
dave.morris@manchester.ac.uk

November 2026 - Where we are now

Harness Engineering

Developing re-usable AGENTS.md, rules and skills.

https://github.com/Zarquan/Acsala-agents

DeepSeek Harness

Open source harness, available on GitHub

https://github.com/deepseek-ai/deepseek-harness

Running the agent in a container.

https://github.com/Zarquan/Lithosia-docker

DeepSeek V4.1 Flash

Open weights model, available on hugging face.

https://huggingface.co/deepseek-ai/DeepSeek-V4.1-Flash-1017

GreenPT inference service

European infrastructure, renewable energy

https://www.greenpt.ai/

Dave Morris
dave.morris@manchester.ac.uk

Thank you

Dave Morris
dave.morris@manchester.ac.uk

GitHub

https://github.com/Zarquan/Lithosia-docker

https://github.com/Zarquan/Acsala-agents

Running to keep up

Please join the discussion.

We need your help to make it happen.

Dave Morris
dave.morris@manchester.ac.uk